

Getayawkal Girma

Addis Ababa | Email |
GitHub | LinkedIn | website

Experience

Lead ERP Developer / Frappe (Python, Java Script)

Habtech Solutions

Oct 2025 – Present

- Led the end-to-end design and implementation of a Construction Project Management System (CPMS) for the Ministry of Health (MOH) using the Frappe framework.
- Architected and customized ERPNext modules to deliver a Primavera-style scheduling and project control system tailored for large-scale construction projects.
- Contributed a bug fix to *Frappe Insights* (open-source) by resolving a permission validation issue where custom fields in child tables were not correctly included in access checks, improving analytics reliability and access control.
- Worked across the full stack: Python (Frappe backend), REST APIs, and Frappe UI components for interactive dashboards and reports.

Backend Engineer / AI Tooling Contributor (FastAPI, LLM Training)

Turing (Contract)

Jan 2025 – Dec 2025

- Collaborated on LLM training projects focused on teaching AI agents to safely and correctly use real-world developer tools.
- Designed and implemented FastAPI-based backend services that simulate real APIs (Stripe, Google Drive, PayPal) inside controlled training environments.
- Built multi-API “gym” environments that allow AI agents to practice tool usage with realistic request/response flows and failure scenarios.
- Implemented strict validation, permissions, and safety constraints to prevent destructive actions during agent training.
- Modeled APIs using Pydantic schemas to enforce structured inputs, outputs, and error handling.
- Created task-oriented endpoints that mirror real developer workflows such as payments, file management, and account operations.
- Worked closely with prompt designers and evaluators to improve agent reliability, correctness, and tool-selection behavior.
- Wrote tests and sanity checks to ensure deterministic behavior of training environments across runs.

Full-Stack Developer (.NET, Angular)

Addis Ababa University

Feb 2025 – Aug 2025

- Developed and deployed full-stack features across five IMIS modules (Research, Workflow, Notice, Memo, Delegation) Management Systems for foreign affairs using Angular and .NET Framework, improving task completion time by 30%.
- Designed and optimized analytics dashboards with SQL star-schema queries, enabling accurate KPIs and performance metrics through Slowly Changing Dimensions.
- Implemented server-side pagination for 100+ records, reducing load time by 45% and improving UI responsiveness.
- Refactored monolithic backend into modular services for maintainability and scalability.

- Enhanced Angular state management with RxJS and Observables, achieving real-time UI synchronization across modules.

Intern (.NET, Angular)

Addis Ababa University

Sep 2024 – Jan 2025

- Adapted to large-scale Angular + .NET codebases, resolving 20+ integration bugs and optimizing API endpoints.
 - Contributed to multilingual support (English/Amharic) by implementing i18n translation features.
 - Built notification and delegation modules to streamline approval workflows across departments.
 - Generated PDF reports via QuestPDF, converting unstructured data into reliable formatted documents.
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Projects

Impostor – Offline-First Social Deduction Game

React Native (Expo), TypeScript, SQLite, Appwrite, Python, LLMs

- Designed and built a cross-platform mobile game using React Native (Expo) with a fully offline-first architecture backed by SQLite.
- Architected a hybrid data strategy combining local SQLite storage with Appwrite (BaaS), enabling seamless offline gameplay with incremental cloud synchronization.
- Implemented an optimized SQL-based content selection algorithm that prioritizes least-used records while maintaining randomness, reducing repetition and improving replayability.
- Built an incremental sync mechanism that fetches only newly verified records from the cloud, minimizing bandwidth usage and cold-start latency.
- Developed an automated Python-based ETL pipeline that scrapes external datasets, deduplicates entries, enriches content using LLMs, and loads verified data into production.
- Integrated AI-powered translation and semantic categorization workflows to support bilingual (English/Amharic) gameplay content.
- Implemented human-in-the-loop verification by gating AI-generated content with review flags before client distribution.

Mezgeb – Offline Receipt Manager

Demo

Ionic, Angular, TypeScript, SQLite, Regex, GitHub CI/CD

- Built an offline-first mobile app that parses and organizes SMS, PDF, and XML bank receipts (CBE, Telebirr, BoA) into a structured SQLite database.
- By storing all transactions locally, Mezgeb can eliminate up to 10 million old receipt server requests per month for a mid-sized bank if fully adopted.
- Implemented 40+ regex patterns for multilingual (English/Amharic) parsing, achieving 98% accuracy on 4,000+ real receipts.
- Engineered ultra-fast query logic capable of loading thousands of receipts in milliseconds, maintaining smooth performance even on low-end Android devices.
- Designed a modular architecture with 9 services and 72 source files (7.8K+ LOC) following clean architecture and dependency injection principles.
- Created interactive analytics dashboards with Chart.js, providing spending insights and visual breakdowns by category.

- Integrated a custom Capacitor native plugin to bypass SSL/CORS issues for PDF parsing, enabling seamless enrichment workflows.
 - Automated updates and releases via GitHub Actions for continuous delivery.
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Education

Bachelor of Mechatronics Engineering Addis Ababa Science and Technology University (AASTU)
Graduated: 2024

Relevant Coursework: Object-Oriented Programming in Python, Machine Learning, Robotics, Computer Vision, Digital Logic Design, Embedded Systems & Instrumentation, Signals & Systems, Integrated Engineering Team Project

Technical Skills

Languages: Python, TypeScript, C#, SQL

Frameworks: FastAPI, Angular, Ionic, .NET, React, React - Native

Databases: SQLite, PostgreSQL, MySQL

Tools: Docker, GitHub Actions, Postman, Regex, QuestPDF, Capacitor

Concepts: Async Programming, REST API Design, CI/CD, Dependency Injection, i18n, Offline-first Systems, AI Integration, TDD